MERFISH Spatial Transcriptomics — Cell-Type Annotation

2nd Prize · University of Rochester Biomedical Data Science Hackathon 2026 · Team XDRKAMOA

Code: github.com/Shomao1998/merfish-celltype-annotation · Competition: github.com/Rochester-Biomedical-DS/Hackathon-Summer-2026

BACKGROUND

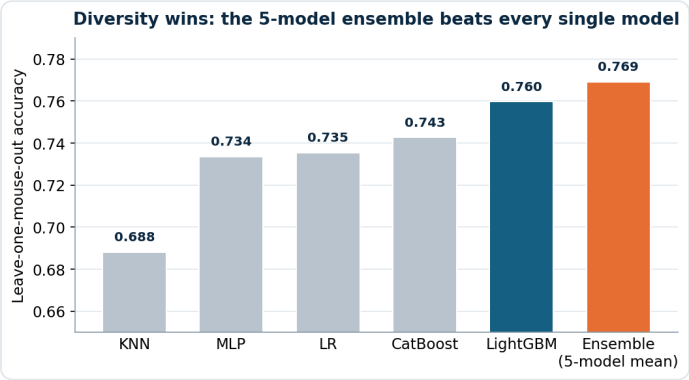
A **60-class cell-type classification** challenge on **MERFISH** spatial-transcriptomics data: predict the cell type of each cell in a mouse spinal-cord dataset (~10,000 cells, **200 genes**), scored by overall accuracy. The data is small, extremely **sparse** (~93% zeros; a median cell has only 12 of 200 genes detected and ~21 transcripts total), and highly **imbalanced** (largest class 703 cells vs. smallest 3).

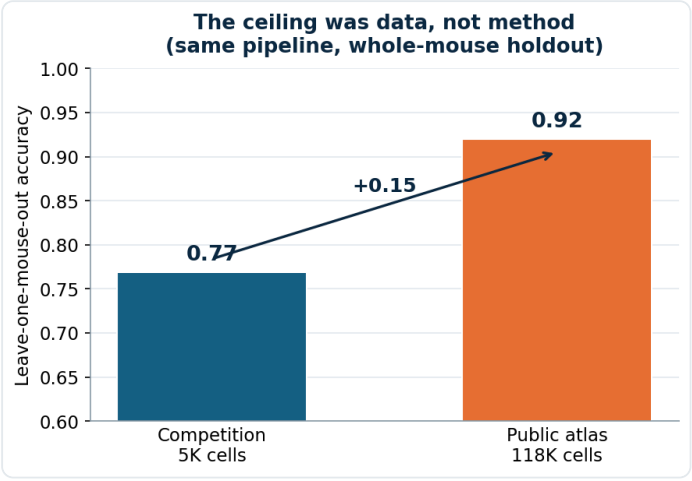
RESULTS

2nd
place — team XDRKAMOA

0.769
leave-one-mouse-out accuracy — final 5-model ensemble (best single model 0.760)

0.77 → 0.92
accuracy as labeled data scales 5K→118K cells (post-competition study)





KEY STEPS

1. **EDA & data dictionary.** Key insight: the 60 types form a clean E/I partition — 24 excitatory / 15 inhibitory / 21 non-neuronal, **zero overlap** — which shaped the decoder.
2. **Biology-constrained decoding.** A hard mask keeps every prediction inside its excitatory/inhibitory group — a whole class of impossible errors removed.
3. **Five-model ensemble.** LightGBM + Logistic Regression + MLP + KNN + CatBoost, fused by equal-probability averaging — **~1 point** over the best single model, purely from diversity.
4. **Fold-safe target encoding.** Anatomical groups → out-of-fold per-group cell-type priors; our single most effective feature on 200 sparse genes.
5. **Generalization-first validation.** Leave-one-mouse-out & leave-one-batch-out GroupKFold, with experiment IDs dropped from features — closed a real **~0.02 CV→leaderboard gap**.

DIRECTIONS WORTH EXPLORING FURTHER

Underperformed on 5K cells but **validation-limited, not wrong** — a roadmap for the 10^5–10^6-cell datasets this field is moving toward.

- **Expression label features** (centroid + kNN vote) — noisy at 5K, stabilize with more reference cells.
- **Anatomical bucket constraint** (Region×E/I×Segment) — a finer E/I mask once bucket coverage fills in.
- **Deep & spatial models** (CNN/DNN, semi-supervised) — data-hungry; should overtake at scale.
- **LLM reranker (Qwen)** — a no-external-data decision-flow interceptor; research only, never submitted.

MY ROLE

Technical lead. Built the **end-to-end pipeline** — from exploratory data analysis and a data dictionary through modeling to final prediction — collaborating with three first-time-competitor teammates from very different backgrounds, and set the validation strategy that drove every modeling decision. Also ran a post-competition data-scaling study and a no-external-data LLM-reranker experiment.

Stack: Python · scikit-learn · LightGBM · CatBoost · pandas · GroupKFold · MERFISH / single-cell spatial transcriptomics.

Integrity: the submitted model uses **zero external data**; we identified — and did not exploit — a leaderboard leak in the public source atlas.